Chapter 16 — Lab Safety and Practical Skills

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. In lab, always wear:

(a) Sandals only **(b)** Lab coat and goggles **(c)** Loose clothes **(d)** No protection

2. Smell a gas by:

(a) Tasting **(b)** Direct inhalation **(c)** Wafting **(d)** Touching

3. Chemicals should never be:

(a) Labelled **(b)** Tasted **(c)** Measured **(d)** Stored

4. Broken glass is a:

(a) Physical hazard **(b)** Biological hazard **(c)** Chemical hazard **(d)** No hazard

5. Concentrated acid is a:

(a) Physical hazard only **(b)** Chemical hazard **(c)** No hazard **(d)** Biological hazard

6. Microorganisms are:

(a) Biological hazard **(b)** Chemical hazard only **(c)** Physical hazard **(d)** No hazard

7. Chemical waste should be:

(a) Evaporated always **(b)** Collected properly **(c)** Poured down sink **(d)** Thrown in trash

8. Acid splash on skin first:

(a) Rub with cloth **(b)** Ignore **(c)** Add base directly **(d)** Wash with water

9. Burette delivers:

(a) Analyte only **(b)** Titrant **(c)** Distillate **(d)** Indicator only

10. Pipette measures:

(a) Indicator pH **(b)** Analyte volume **(c)** Titrant mass **(d)** Temperature

11. Indicator shows:

(a) End point **(b)** Boiling point **(c)** Molar mass **(d)** Density

12. Concordant titres are:

(a) One reading only **(b)** Zero readings **(c)** Random readings **(d)** Consistent readings

13. Meniscus read at:

(a) Below **(b)** Above **(c)** Any angle **(d)** Eye level

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14. Carbonate with acid gives:

(a) CO_2 **(b)** O_2 **(c)** N_2 **(d)** H_2

15. CO_2 turns limewater:

(a) Colourless always **(b)** Blue **(c)** Milky **(d)** Red

16. Sulfate test uses:

(a) $AgNO_3$ **(b)** NaOH only **(c)** HCl only **(d)** $BaCl_2$

17. Sulfate precipitate is:

(a) $BaSO_4$ **(b)** AgCl **(c)** $CaCO_3$ **(d)** Fe(OH)3

18. Chloride test uses:

(a) H_2SO_4 only **(b)** $AgNO_3$ **(c)** $BaCl_2$ **(d)** NaOH only

19. Nitrate confirmatory test:

(a) $BaCl_2$ **(b)** Limewater **(c)** Flame test **(d)** Brown ring

20. Al_3+ with NaOH gives:

(a) Blue ppt **(b)** White ppt soluble excess **(c)** Green ppt **(d)** No ppt

21. Zn_2+ with excess ammonia:

(a) Gas **(b)** Insoluble **(c)** Soluble **(d)** No change

22. Cu_2+ with excess ammonia gives:

(a) No colour **(b)** White ppt **(c)** Deep blue solution **(d)** Green ppt

23. Fe_2+ with NaOH gives:

(a) Green ppt **(b)** No ppt **(c)** Red-brown ppt **(d)** Blue solution

24. Fe_3+ with NaOH gives:

(a) White ppt **(b)** Red-brown ppt **(c)** No ppt **(d)** Green ppt

25. NH_4^+ with NaOH on warming gives:

(a) N_2 **(b)** NH_3 **(c)** NO_2 **(d)** H_2

26. Na^+ flame colour:

(a) Golden yellow **(b)** Lilac **(c)** Brick red **(d)** Blue

27. K^+ flame colour:

(a) Green **(b)** Red **(c)** Yellow **(d)** Lilac

28. Ca_2+ flame colour:

(a) Brick red **(b)** Lilac **(c)** Yellow **(d)** Blue

29. Standard solution has:

(a) No solute **(b)** Known concentration **(c)** Unknown concentration **(d)** Only indicator

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30. End point is shown by:

(a) Temperature only **(b)** Gas pressure **(c)** Indicator colour change **(d)** Mass loss

31. Laboratory coat protects:

(a) Hair only **(b)** Shoes only **(c)** Skin and clothing **(d)** Eyes only

32. Goggles protect:

(a) Hands **(b)** Eyes **(c)** Feet **(d)** Hair

33. Physical hazard example:

(a) Bacteria **(b)** Toxic vapour **(c)** Acid burn **(d)** Broken glass

34. Chemical hazard example:

(a) Broken glass **(b)** Corrosive acid **(c)** Hot plate only **(d)** Electric shock

35. Biological hazard example:

(a) Broken glass **(b)** Flame **(c)** Microorganisms **(d)** Electricity

36. Chemical waste should be:

(a) Collected in labelled containers **(b)** Mixed together **(c)** Evaporated always **(d)** Poured randomly

37. Acid splash first aid:

(a) Wipe dry only **(b)** Ignore **(c)** Wash with plenty water **(d)** Neutralize with strong base

38. Titration uses burette to deliver:

(a) Analyte only **(b)** Solvent only **(c)** Indicator only **(d)** Titrant

39. Pipette measures:

(a) Fixed volume of analyte **(b)** Variable mass **(c)** Temperature **(d)** Pressure

40. Burette rinsed with:

(a) Distilled water only **(b)** Solution it will contain **(c)** Acid always **(d)** Base always

41. Concordant titres are:

(a) Always zero **(b)** Different by 10 mL **(c)** Close agreeing readings **(d)** Random readings

42. Carbonate with acid gives:

(a) CO_2 **(b)** H_2 **(c)** N_2 **(d)** O_2

43. Sulfate with $BaCl_2$ gives:

(a) Yellow AgI **(b)** White $BaSO_4$ **(c)** White AgCl **(d)** Blue Cu(OH)2

44. Chloride with $AgNO_3$ gives:

(a) Cream AgBr **(b)** Yellow AgI **(c)** White AgCl **(d)** Black Ag_2S

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45. Nitrate confirmatory test:

(a) Flame test **(b)** Brown ring test **(c)** Biuret test **(d)** Limewater test

46. Al_3+ with NaOH gives:

(a) Red-brown ppt **(b)** Blue ppt **(c)** White ppt soluble in excess **(d)** Green ppt

47. Zn_2+ with excess NH_3 gives:

(a) Red-brown ppt **(b)** Green ppt **(c)** Soluble complex **(d)** Insoluble ppt always

48. Cu_2+ with excess NH_3 gives:

(a) Colourless solution **(b)** Deep blue solution **(c)** Green ppt only **(d)** White ppt

49. Fe_3+ with NaOH gives:

(a) Red-brown ppt **(b)** White ppt **(c)** Green ppt **(d)** Blue solution

50. NH_4^+ with NaOH on warming gives:

(a) N_2 **(b)** NO_2 **(c)** H_2 **(d)** NH_3

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Answer Key with Brief Reasons

1. **Answer: (b) Lab coat and goggles** — Safety rule.
2. **Answer: (c) Wafting** — Safe technique.
3. **Answer: (b) Tasted** — Safety rule.
4. **Answer: (a) Physical hazard** — Hazard type.
5. **Answer: (b) Chemical hazard** — Corrosive.
6. **Answer: (a) Biological hazard** — Lab hazard.
7. **Answer: (b) Collected properly** — Safe disposal.
8. **Answer: (d) Wash with water** — First aid.
9. **Answer: (b) Titrant** — Titration.
10. **Answer: (b) Analyte volume** — Titration.
11. **Answer: (a) End point** — Colour change.
12. **Answer: (d) Consistent readings** — Reliability.
13. **Answer: (d) Eye level** — Avoid parallax.
14. **Answer: (a) CO_2** — Gas test.
15. **Answer: (c) Milky** — $CaCO_3$ precipitate.
16. **Answer: (d) $BaCl_2$** — $BaSO_4$ white ppt.
17. **Answer: (a) $BaSO_4$** — White ppt.
18. **Answer: (b) $AgNO_3$** — AgCl white ppt.
19. **Answer: (d) Brown ring** — NO_3- test.
20. **Answer: (b) White ppt soluble excess** — Amphoteric hydroxide.
21. **Answer: (c) Soluble** — Complex formation.
22. **Answer: (c) Deep blue solution** — Complex ion.
23. **Answer: (a) Green ppt** — Fe(OH)2.
24. **Answer: (b) Red-brown ppt** — Fe(OH)3.
25. **Answer: (b) NH_3** — Ammonia gas.
26. **Answer: (a) Golden yellow** — Flame test.
27. **Answer: (d) Lilac** — Flame test.
28. **Answer: (a) Brick red** — Flame test.
29. **Answer: (b) Known concentration** — Titration.
30. **Answer: (c) Indicator colour change** — Titration.
31. **Answer: (c) Skin and clothing** — PPE.

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32. Answer: (b) Eyes — Splash protection.

33. Answer: (d) Broken glass — Energy/object hazard.

34. Answer: (b) Corrosive acid — Chemical property.

35. Answer: (c) Microorganisms — Living agents.

36. Answer: (a) Collected in labelled containers — Safe disposal.

37. Answer: (c) Wash with plenty water — Immediate dilution.

38. Answer: (d) Titrant — Known solution.

39. Answer: (a) Fixed volume of analyte — Accurate volume.

40. Answer: (b) Solution it will contain — Avoid dilution.

41. Answer: (c) Close agreeing readings — Reproducibility.

42. Answer: (a) CO_2 — Limewater test.

43. Answer: (b) White $BaSO_4$ — Acidic medium.

44. Answer: (c) White AgCl — Soluble in dilute NH_3.

45. Answer: (b) Brown ring test — $NO\text{-}Fe_2+$ complex.

46. Answer: (c) White ppt soluble in excess — Amphoteric hydroxide.

47. Answer: (c) Soluble complex — Ammine complex.

48. Answer: (b) Deep blue solution — Tetraammine complex.

49. Answer: (a) Red-brown ppt — Fe(OH)3.

50. Answer: (d) NH_3 — Turns red litmus blue.

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